

**REPORT FROM MEMBER STATE THE NETHERLANDS ON THE IMPLEMENTATION OF THE
2021 RECOMMENDATION ON OPEN SCIENCE**

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- **Country*:** The Netherlands
- **Organization(s) or entity(ies) responsible for the preparation of the report*:**
Name: Ministry of Education, Culture and Science,
Department for Research and Science Policy
Website: www.government.nl/ministries/ministry-of-education-culture-and-science
Email address*:
Phone number*:
Please describe the role/mandate of your organization:

The Ministry of Education, Culture and Science¹ is a central government organization responsible for national policies regarding education, culture, science (including open science) and equal opportunities.

Mission statement: The Ministry of Education, Culture and Science works to create a smart, skilled and creative environment in the Netherlands. Its mission is to ensure that everyone gets a good education and is prepared for responsibility and independence. The Ministry also wants people to enjoy the arts, and aims to create the right conditions for teachers, artists and researchers to do their work. Our mission is to ensure a knowledgeable, skilled and cultured country.

The Ministry has the following objectives:

- To ensure that everyone gets a good education
- To ensure that everyone is prepared for personal independence and responsibility
- To ensure that everyone has the opportunity to experience and enjoy culture
- To ensure that teachers, artists and scientists are able to carry out their work
- **Officially designated contact person(s) who completed this survey:**
Full name*: Mrs. Gerdien Verheuvél
Position*: Senior Policy Adviser
Email address*:
- **Full Name(s) of designated official(s) certifying the report:**
Mr. Bert Hiemstra, Management Team Member Department for Research and Science Policy
- **Brief description of the consultation process established for the preparation of the report:** This report is drafted in consultation with key open science partners in the Netherlands, notably Open Science NL and the Netherlands National Commission for UNESCO.
- **Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:**
Name of the organization*: Open Science NL, part of the Dutch Research Council NWO
Website*: www.openscience.nl/en / www.nwo.nl/en/open-science
Email address*:
Sector: ☒ Public ☐ Private ☐ Civil Society ☐ Other
Name of the organization*: Netherlands National Commission for UNESCO
Website*: www.unesco.nl/nl/dossier/open-science
Email address*:
Sector: ☒ Public ☐ Private ☐ Civil Society ☐ Other

B. Questionnaire for Member States' reporting on the implementation of the 2021 Recommendation

Disclaimer: Open science has been a national policy priority in the Netherlands since 2016 and there is an active open science community. Open science has been actively practiced and promoted by the various open science stakeholders at various levels (e.g. national, funding, umbrella organisations, institutional, communal, individual) in the country for at least a decade. This has resulted in many (national and institutional) policies, initiatives and events over the years. The answers below contain some of the main ones and should only be considered as a selection of the numerous initiatives taken over the years.

¹ www.government.nl/ministries/ministry-of-education-culture-and-science

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES / NO *[If yes, please indicate which ministries and/or national authorities/entities have been involved in the promotion of the 2021 Recommendation.]*

The 2021 Recommendation has been shared with:

- Ministries: Ministry of Economic Affairs; Ministry of Health, Welfare and Sport; Ministry of the Interior and Kingdom Relations; Ministry of Justice and Security.
- National authorities/entities: the main open science stakeholders in the Netherlands, e.g. Open Science NL (the 10 year national initiative for Open Science in the Netherlands embedded within the Dutch Research Council NWO) and all affiliated stakeholders², Nationale UNESCO Commissie, and others.

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES / NO

1.3 Have there been awareness raising activities on open science, including all its key elements, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: [\(i\) 16.f, 16.g, 16.h](#))

YES / NO *[If yes, please provide details on the activities organized or foreseen and links as relevant.]*

Please note that the following list gives an impression of some main awareness raising activities, but is non-exhaustive.

➤ **Important national documents and guidelines** that feature UNESCO's *Recommendation on Open Science* and its definition of Open Science as guiding principles:

- "Open Science 2030 in the Netherlands" Ambition Document (2022)
www.openscience.nl/en/the-assignment-of-open-science-nl
www.openscience.nl/sites/open_science/files/media-iles/final_npos2030_ambition_document_and_rolling_agenda.pdf
www.rijksoverheid.nl/documenten/rapporten/2022/12/07/bijlage-npos-2030-ambitiedocument-en-rolling-agenda
- Covenant Regieorgaan Open Science NL (2023)
www.openscience.nl/en/the-assignment-of-open-science-nl
www.openscience.nl/sites/open_science/files/media-files/Convenant%20regieorgaan%20Open%20Science.pdf
- Open Science NL Work Programme 2024-2025
www.openscience.nl/en/news/open-science-nl-presents-work-programme-for-2024-and-2025
www.openscience.nl/sites/open_science/files/media-files/Work%20programme%202024-2025_0.pdf
- Open Science Agenda Universities of the Netherlands (2023)
www.universiteitenvannederland.nl/files/publications/Open%20Science%20agenda%20UNL.pdf
- Netherlands eScience Center's Practical guide to Software Management Plans (2022)
www.esciencecenter.nl/news/the-practical-guide-to-software-management-plans-now-available/
- Royal Netherlands Academy of Arts and Sciences (KNAW): Publication `Science Communication by Scientists: Rewarded! (2022)
www.knaw.nl/en/publications/science-communication-scientists-rewarded
- Citizen Science Nederland (CS-NL) vision document (2024)
<https://citizenscience.nl/>

➤ **Awareness raising events and activities:**

A selection of some of the main events and activities (among the many in the country):

- annual Open Science Festival (yearly, started in 2021) <https://opensciencefestival.nl/>
- annual Recognition and Rewards Festival (yearly, started in 2021)
<https://recognitionrewards.nl/festival/recognition-rewards-festival-2024/>
- annual Citizen Science Nederland National Networking Day (started in 2022)
<https://citizenscience.nl/p/jaarlijkse-netwerkdag/> / <https://samenmetenutrecht.nl/citizen-science-nederland-netwerkdag-2025/>
- incentive for open science activities: Open Science NL call for Open Science meetings 2025
www.openscience.nl/en/calls/open-science-meetings-2025

² The affiliated stakeholders to Open Science NL are : NWO, UNL, NFI, VRI, SURF, KNAW, NWO-I, ZonMw, UKB, KB, DANS, Netherlands eScience Center, 4TUResearchData, Health-RI en SHB and the Ministry of Education, Culture and Science.

- community consultations on open science priorities and challenges through workshops and advisory panels, which are regularly hosted by Open Science NL and described in the Open Science NL's Community engagement strategy:
www.openscience.nl/sites/open_science/files/media-files/Open%20Science%20NL%20Community%20engagement%20strategy_0.pdf
- Recognition and Rewards toolkit for dialogue. (to support those who wish to organise a dialogue on Recognition and Rewards in their institution)
<https://recognitionrewards.nl/practices/toolkit-for-dialogue/>

Webpages of two awareness raising communities:

- Open Science Communities-NL: <https://osc-international.com/open-science-community-the-netherlands/>
- Citizen Science-NL: <https://www.cs-nl.network/>

Activities by the Netherlands' National Commission for UNESCO:

- Articles on the website:
2021: www.unesco.nl/nl/artikel/unesco-neemt-aanbevelingen-aan-over-open-science-en-over-ethiek-van-ai
2022: www.unesco.nl/nl/artikel/volgende-stap-voor-het-implementeren-van-de-aanbeveling-over-open-science
- Prof. Sarah de Rijcke, member of the Open Science Advisory Committee for the UNESCO Open Science Recommendation 2021, was in 2022 appointed by the Minister of Education, Culture and Science as member of the Netherlands' National Commission for UNESCO in December 2022.
www.unesco.nl/nl/medewerker/sarah-de-rijcke
www.unesco.nl/nl/artikel/sarah-de-rijcke-en-jeroen-geurts-treden-toe-tot-de-unesco-commissie
- Brochure on the Open Science Recommendation (23-04-2024)
www.unesco.nl/nl/artikel/unesco-commissie-lanceert-brochure-over-open-science
- Keynote speech at EuroScience Open Forum (2022):
The Chair of the Commission was key note speaker at ESOE 2022, the largest multidisciplinary scientific congress of Europe, where she spoke of UNESCO's Open Science mission.
- Dissemination of the UNESCO brochure on the Open Science Recommendation (2024):
www.unesco.nl/nl/artikel/unesco-commissie-lanceert-brochure-over-open-science
The Brochure was sent to 220 relevant (international) Open Science contacts of the Commission. 43 stakeholders within the Netherlands received a physical copy, alongside a formal letter.
- Reflection on the Open Science NL work program (in Dutch and English, 2024):
The Commission, together with the UNESCO Chair in Diversity and Inclusion in Global Science at Leiden University, has assessed the national work program on Open Science by viewing it through a 'UNESCO lens'. The reflection was also shared with 224 relevant (international) Open Science contacts of the Commission.
www.unesco.nl/nl/artikel/unesco-commissie-reflecteert-op-werkprogramma-van-open-science-nl
- Workshop on AI and Open Science (2024):
www.unesco.nl/nl/event/meld-je-nu-aan-unesco-workshop-over-ai-en-open-science
A workshop on how AI can help bring science closer to the general public, including presentations by citizen scientist projects and a panel with relevant societal stakeholders. In collaboration with the UNESCO Chair in Diversity and Inclusion in Global Science at Leiden University.
- Open Science Monitoring (2024):
The Commission's advisor presented the monitoring process for UNESCO's Open Science Recommendation at an expert's meeting on Open Science monitoring at TU Delft :
www.tudelft.nl/en/events/2024/open-science/diverse-approaches-to-monitoring-open-science

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science, in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

YES / NO [If yes, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.]

The 2021 UNESCO Open Science Recommendation is explicitly mentioned in the main, most recent (national) Open Science documents (e.g. the 'Open Science 2023 in the Netherlands: NPOS Ambition Document and Rolling Agenda'³, and in the Open Science NL work program 2024-2025⁴). The main open science stakeholders in The Netherlands have committed themselves to these documents and as such to incorporating the values and principles of open science in publicly funded research.

³ 'Open Science 2023 in the Netherlands: NPOS Ambition Document and Rolling Agenda [published on 13-12-2022] <https://doi.org/10.5281/zenodo.7433767>

⁴ Open Science NL work program 2024-2025:
www.openscience.nl/en/news/open-science-nl-presents-work-programme-for-2024-and-2025

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy strategy or plan of action on science, technology and innovation (STI)?

YES / NO [If yes, please attach/upload a machine-readable PDF file of the policy and provide the following information]

Brief context: The Dutch Research Council NWO and several academic institutions and other stakeholders in the Netherlands have been early adopters of open science. At national and European level open science gained momentum when the Netherlands put open science on the agenda of the Netherlands' Presidency of the Council of the European Union in 2016. Under this Presidency, Council Conclusions on the transition towards an open science system⁵ were adopted and the Amsterdam Call for Action on Open Science⁶ was launched. Since then, the Netherlands' government, in the STI paragraphs of two consecutive coalition agreements (2016-2021⁷ and 2021-2025⁸), has stated that open science should become the norm in scientific research. Under the current government (2024-2028⁹) the existing open science policies are being continued.

Several national policy letters, strategies and actions on STI have followed from these coalition agreements. The two main overarching science and research policy letters (from 2019¹⁰ and 2022), from which the current open science policies followed, are mentioned below.

- **Title of the policy:** Higher Education and Science Policy
- **Authority in charge of development of the policy:**
Ministry of Education, Culture and Science
(please note that science and research policy, including open science, is under the responsibility of the Ministry of Education, Culture and Science. The Ministry of Economic Affairs is responsible for Technology and Innovation policy. The two ministries work closely together.)
- **Entities involved and the process of development:**
All relevant stakeholder organisations in the Netherlands
- **Year of adoption:** 2022
- **Key areas/sections of the policy** major science and research policy topics are described along the lines of 3 objectives:
 - Objective 1: Strong and healthy foundations
 - Objective 2: Creating scope for a wide range of talent
 - Objective 3: Increasing the societal impact of higher education and research [open science is part of this section of the policy letter]
- **Links to the webpage**
www.government.nl/documents/parliamentary-documents/2022/06/17/letter-to-parliament-about-higher-education-and-science-policy

- **does it include any of the following open science elements:** Please note that the open science section of the afore mentioned policy letter is only brief. More detailed open science dedicated policy letters, documents and initiatives followed from this letter.
 - **open access to scientific knowledge:** Yes
 - **open science infrastructure:** Yes
 - **open engagement of societal actors:** Yes
 - **open dialogue with other knowledge systems:** Not specifically mentioned in this concise policy letter.

⁵ <https://data.consilium.europa.eu/doc/document/ST-9526-2016-INIT/en/pdf>

⁶ https://en.wikipedia.org/wiki/Amsterdam_Call_for_Action_on_Open_Science

⁷ www.rijksoverheid.nl/documenten/publicaties/2017/10/10/regeerakkoord-2017-vertrouwen-in-de-toekomst

⁸ www.rijksoverheid.nl/documenten/publicaties/2022/01/10/coalitieakkoord-omzien-naar-elkaar-vooruitkijken-naar-de-toekomst

⁹ www.rijksoverheid.nl/regering/regeerprogramma

¹⁰ www.rijksoverheid.nl/documenten/kamerstukken/2019/01/28/aanbiedingsbrief-bij-wetenschapsbrief-nieuwsgierig-en-betrokken-de-waarde-van-wetenschap

2.2 In your country, is there a policy, or a set of policies and/or legal framework(s) at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) [16.a](#), [16.b](#), [16.c](#), [16.d](#), [16.e](#), [16.h](#))

YES / PARTLY / NO [If yes, please attach/upload machine-readable **PDF** file(s) of the policy(ies) and provide the following information for each:]

Following from the under 2.1. mentioned umbrella document, open science policy has since 2013 laid down in various interlinked **policy documents**, that are in line with the 2021 Recommendation on Open Science, e.g. (in order of appearance):

The open science policy documents contain the main open science pillars:

- Open access to scientific publications (open scholarly communication)
- FAIR data and open research software
- Citizen science and societal engagement
- Recognition and rewards of researchers

Policy documents and links to webpages (non-exhaustive list):

Letters from the Minister of Education Culture and Science to Parliament that were all drafted in close contact and cooperation with main open science stakeholders in the Netherlands:

- Letter on open access to publications [15-11-2013]
www.rijksoverheid.nl/documenten/kamerstukken/2013/11/15/kamerbrief-over-open-access-van-publicaties
- Letter on the progress regarding open science [19-01-2017]
www.rijksoverheid.nl/documenten/kamerstukken/2017/01/19/kamerbrief-over-voortgang-open-science
- Letter 'curious and committed, the value of science' [28-01-2019]
www.rijksoverheid.nl/documenten/kamerstukken/2019/01/28/aanbiedingsbrief-bij-wetenschapsbrief-nieuwsgierig-en-betrokken-de-waarde-van-wetenschap
- Letter on Rathenau Institute's factsheet on open science [22-02-2019]
www.tweedekamer.nl/kamerstukken/brieven_regering/detail?id=2019Z03699&did=2019D07870
- Letter on the progress on open science of scientific publications [05-11-2019]
www.tweedekamer.nl/kamerstukken/brieven_regering/detail?id=2019Z21226&did=2019D44221
- Letter on cOAlition S and Plan S [09-10-2020]
www.tweedekamer.nl/kamerstukken/brieven_regering/detail?id=2020Z18449&did=2020D39876
- Letter 'Open Science NL accelerates the transition to open science' (on the covenant open science, including the establishment of the 10-year program Open Science NL [29-03-2023] **PDF added**
www.rijksoverheid.nl/documenten/kamerstukken/2023/03/29/open-science-nl-versnelt-transitie-naar-open-science
- Letter on Higher Education and Research [17-06-2022, pages 25-26] **PDF added**
www.rijksoverheid.nl/documenten/kamerstukken/2022/06/17/aan-de-tweede-kamer-beleidsbrief-hoger-onderwijs-en-wetenschap
- Letter on progress of the Recognition and Rewards Programme [21-06-2024] **PDF in added**
www.tweedekamer.nl/kamerstukken/brieven_regering/detail?id=2024Z11107&did=2024D26476

- **Title of the legal instrument:**
The Taverne Amendment (article 25fa), part of the Netherlands' Copyright Law
- **Authority in charge of development of the policy:**
Ministry of Justice and Safety and Ministry of Education, Culture and Science
- **Entities involved and the process of development:**
Various stakeholders from the public and private sector
- **Year of adoption of the Taverne Amendment:** 2015
- **Year of last update:** the umbrella Copyright Law was revised in 2024, but article 25fa was left unchanged in that process
- **Key areas/sections of the policy / Elements of open science that are addressed:** open access to scientific publications
- **Links to webpage(s)**
 - Dutch Copyright Act - Article 25fa (Amendment Taverne):
<https://zoek.officielebekendmakingen.nl/kst-33308-11> **PDF added**
 - Articles on the pilot 'You share, we take care' on sharing published short academic works in institutional repositories after six months :
<https://doi.org/10.53377/lq.10915> **PDF added**
www.tudelft.nl/en/library/support/library-for-researchers/publishing-outreach/you-share-we-take-care
www.universiteitiden.nl/en/news/2019/01/dutch-universities-give-open-access-another-boost
www.uu.nl/en/achtergrond/make-publications-open-access-available
www.openaccess.nl/sites/default/files/documents/2025-02/summary_taverne_pilot_vsnu_200330.pdf

The Taverne amendment is a provision for Secondary Publishing Rights laid down in Article 25fa of the Copyright Act. It offers the (personal) right to researchers in the Netherlands to freely share their publicly funded research outputs by means of a repository, following a 'reasonable period of time'. Via the pilot project 'You share, we take care' in 2019, the universities of the Netherlands have defined this 'reasonable period of time' as 6 months for the Version of Record, and that eligible outputs are articles and book chapters. All universities have developed workflows which help researchers to make their output openly available via this route. Many of them have adopted opt-out policies, meaning that publications by authors affiliated with Dutch universities are automatically deposited in the institutional repositories and made available unless a researcher decides to opt out. The policy is highly effective, resulting in almost 100% open access rates for many Dutch institutions. The Dutch Research Council (NWO) is a Plan S signatory, which means the cOAlition S Rights Retention Strategy is an eligible route.

2.3 In your country, are there specific policy instruments that aim to promote open science?

YES / UNDER DEVELOPMENT / NO

[If yes, please provide details and links as relevant, including for each instrument:]

- **Title of the policy instrument:** Open Science NL, *national initiative for Open Science*
- **Authority in charge of development of the policy:** NWO (Dutch Science Council)
- **Entities involved and the process of development:**
 - NWO (Dutch Research Council)
 - Universities NL (UNL)
 - Netherlands' Federation of Academic Medical Centres (NFU)
 - Association of Universities of Applied Sciences (VH),
 - SURF (IT cooperative of education and research),
 - Royal Netherlands Academy of Arts and Sciences (KNAW)
 - NWO-I (Institutes Organisation of NWO)
 - ZonMw
(national funding organisation for research and innovation in health, healthcare and well-being)
 - UKB (joint venture of the 13 Dutch research universities libraries)
 - National Library of the Netherlands (KB)
 - Data Archiving and Networked Services (DANS)
 - Netherlands eScience Center
 - 4TUResearchData
 - Health-RI (enabling data driven health and life sciences)
 - Knowledge Network of University of Applied Sciences Libraries (SHB)
 - Ministry of Education, Culture and Science (OCW)
- **Year of adoption** 2023
- **Key areas/sections of the policy / Elements of open science that are addressed**
Open Science NL works along four program lines:
 - Open scholarly communication
 - FAIR data
 - Open Research Software
 - Citizen science/societal engagement
- **Links to webpage(s)**
www.openscience.nl/
www.nwo.nl/open-science
www.nwo.nl/nieuws/regieorgaan-open-science-officieel-van-start-onder-de-naam-open-science-nl

2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation?
(ref.: [\(ii\) 17.a](#), [17.b](#), [17.c](#), [17.f](#), [17.h](#))

YES / UNDER DEVELOPMENT / NO *[If yes, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.]*

- National Plan(s) on Open Science:
 - Open Science 2023 in the Netherlands: NPOS Ambition Document and Rolling Agenda¹¹ [published 13/12/2022] <https://doi.org/10.5281/zenodo.7433767>
 - Please note that in the UNESCO Open Science Recommendation (2021) is specifically mentioned and build upon in this document.
 - Open Science NL work program 2024-2025: www.openscience.nl/en/news/open-science-nl-presents-work-programme-for-2024-and-2025
 - Please note that the National UNESCO Commission in the Netherlands has written a reflection on this document: www.unesco.nl/nl/artikel/unesco-commissie-reflecteert-op-werkprogramma-van-open-science-nl
 - Position Paper 'Room for Everyone's Talent' regarding recognition and rewards of scientist (research assessment). www.nwo.nl/en/position-paper-room-for-everyones-talent Since 2019 there is a separate national program for recognition and rewards: <https://recognitionrewards.nl/>)

Important weblinks for context on these national plans and strategies:

- Open science NL: www.openscience.nl/en
- NWO (Dutch Research Council):
 - Open Science: www.nwo.nl/en/open-science and
 - Recognition and Rewards: www.nwo.nl/en/recognition-and-rewards
- Universities of the Netherlands (UNL)
 - UNL Open Science Agenda
 - www.universiteitenvannederland.nl/files/publications/UNL_Open%20Science%20agenda%20UNL-EN.pdf
 - www.universiteitenvannederland.nl/onderwerpen/onderzoek/open-access
 - www.universiteitenvannederland.nl/onderwerpen/onderzoek/naar-open-science-als-norm
- Open science vision statement of the universities of applied sciences: www.vereniginghogescholen.nl/themas/open-science-in-het-hbo

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at institutional level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

YES / UNDER DEVELOPMENT/ NO [If yes, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science **Fout! Bladwijzer niet gedefinieerd.** included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).]

This follows from all the aforementioned policies and initiatives. All publicly funded institutions in the Netherlands are involved (either via their umbrella organisations or directly) in the implementation of open science at institutional level.

There are too many policies to count and to mention exhaustively, but here are some weblinks with information and examples to give an impression:

Open Science at the 14 publicly funded research universities in the Netherlands:

- Utrecht University: www.uu.nl/en/research/open-science
- UvA Amsterdam: <https://arils.uva.nl/open-science/open-science.html>
- Free University Amsterdam (VU): <https://vu.nl/nl/over-de-vu/organisaties/open-science>
- TUDelft: www.tudelft.nl/en/open-science
- TUEindhoven: www.tue.nl/en/our-university/calendar-and-events/17-09-2024-everything-you-wanted-to-know-about-open-science-at-tue
- Groningen University: www.rug.nl/research/openscience/
- Leiden University: www.library.universiteitleiden.nl/researchers/open-science
www.universiteitleiden.nl/dossiers/academia-in-motion
- Maastricht University: www.maastrichtuniversity.nl/research/open-science
- Radboud University Nijmegen:

¹¹ The NPOS Ambition Document and Rolling Agenda substitutes the first Netherlands' National Plan on Open science that was published on 9 February 2019 www.openscience.nl/binaries/content/assets/subsites-evenementen/open-science/nationaal_plan_open_science_the_netherlands_february_2017_nl_.pdf

www.ru.nl/medewerkers/interne-projecten/radboud-open-science-programma

- Erasmus University Rotterdam:
www.eur.nl/en/eqsh/course/open-science-and-research-transparency#:~:text=Erasmus%20University%20Rotterdam%20promotes%20open,understanding%20of%20open%20science%20practices.
www.erim.eur.nl/research-integrity/open-science-initiative/
- Tilburg University: www.tilburguniversity.edu/intranet/research-support-portal/open-science-framework
- University of Twente: www.utwente.nl/en/service-portal/research-support/open-science
- Wageningen University and research:
www.wur.nl/en/about-wur/our-values/finding-answers-together/open-science-education.htm?_gl=1*dc131e*_up*MQ..*_ga*MTI2MTE0MDgyMi4xNzQwMDUzODky*_ga_SZ1PQDSPJD*MTc0MDA1Mzg5MS4xLjAuMTc0MDA1Mzg5MS4wLjAuMA
- Open University: www.ou.nl/open-science

Also, there are **Open Science Communities** in each research university city in the Netherlands. These are bottom up initiatives by researchers to enhance and promote open science practices among their peers and institutions. Twelve Communities have received financial support from Open Science NL for their professional development. <https://osc-international.com/open-science-community-the-netherlands/>

Open Science at Universities of Applied Sciences, some information and examples:

- Hogeschool van Amsterdam: www.hva.nl/onderzoek/open-science
- Hogeschool Rotterdam:
www.hogeschoolrotterdam.nl/voorlichting/begeleiding-en-voorzieningen/mediatheek/support/research-support/overzicht/themas/open-science/

2.6 In your country, are there specific funding mechanisms for open science at the national level? (ref.: (ii) 17.i, (iii) 18.i, (iv) 19.c)

YES / UNDER DEVELOPMENT / NO [If yes, please provide details and links as relevant.]

- In the current (and previous) administrative agreement between the Ministry of Education, Culture and Science and the universities regarding the allocation of the lump sum financial resources, it is agreed that 'the institutions will continue to make open science and open education the norm because, in addition to knowledge utilization and impact, this also leads to better quality education and research'.¹²
- In addition, the Ministry of Education, Culture and Science provides funding (80 million euro in total) for the Open Science NL programme, which is the national initiative for Open Science in the Netherlands, running a 10-year (2023-2031) funding program to make open science the norm in Dutch research by 2031. Open Science NL uses these funds to stimulate various aspects of the Open Science agenda. The current work plan period funding is aimed at the following priority areas:
 1. Capacity building,
 2. Infrastructure for Open Science,
 3. Robust research processes,
 4. Evidence base for Open Science,
 5. Empowering communities.

Under these areas a wide range of calls are being opened to addresses a comprehensive set of needs and challenges relating to open science, from supporting from strengthening and supporting local and thematic digital competence centres, national training for data stewardship, open science infrastructure, citizen science hubs, institutional (diamond) open access, open and sustainable research software, the recognition and rewards of open science, replication studies and research into open research.

www.openscience.nl/calls
- In 2020 NWO launched the Open Science Fund that aims to support projects specifically designed to implement and stimulate open science practices. With this funding instrument (grants) NWO has taken a step forward towards changing the way academics are recognised and rewarded in the Netherlands. There was a second round of funding in 2022-2023. www.nwo.nl/en/researchprogrammes/open-science/open-science-fund

¹² www.rijksoverheid.nl/documenten/kamerstukken/2022/07/14/bestuursakkoord-2022-hoger-onderwijs-en-wetenschap. Paragraph 1.6, Page 10.

2.7 In your country, have open science practices been integrated in existing research funding mechanisms?

YES / NO *[If yes, please provide details and links as relevant.] Examples:*

- Since January 2021 the principles of Plan S, the initiative of European research funders to accelerate the transition to open access, have been applying to all NWO/ZonMw calls since that date.¹³
- All NWO grantees are required to make their publications available Open Access and their research datasets as open and FAIR (Findable, Accessible, Interoperable and Reusable) as possible. Furthermore, all grantees are required to submit a data management plan reviewed by a data steward before the start of the project.
- The Dutch Research Council (NWO) has signed DORA in 2019 and has implemented in principles across all its research funding mechanisms. This means three actions in practice:
 1. References to Journal Impact Factors and the h-index have been banned in all call texts and application forms;
 2. Applicants, referees and committee members are actively informed about the policy changes that follow from the signing of DORA;
 3. NWO has introduced the evidence-based CV (formerly a narrative CV).

2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

YES / NO *[If yes, please provide details and links as relevant.]*

For about a decade the Universities in the Netherlands (UNL), the Netherlands Federation of University Medical Centres (NFU) and the Dutch Research Council (NWO) have been negotiating (transformative) 'read and publish' agreements with scientific publishers. The overview of current agreements can be found here: www.openaccess.nl/en/publishing/publisher-deals. Efforts have been made to foster the equitability of these agreements and cooperation with publishers, e.g. initiatives worth mentioning:

- Seven Guiding Principles for Open Research Information (2022): a document for research organisations to adopt when creating and procuring services to manage research information. It was drafted to address issues around the responsible use of research information and the role of commercial third party providers in particular. [<https://zenodo.org/records/6074944>]
- Conference (2024) to explore the successes and challenges of the 'read and publish' deals; cooperation with publishers in the field of research information and how concepts such as 'digital sovereignty' play out in relationships with commercial publishers. An open forum to discuss what the academic sector wants from the publishing sector in safeguarding academic values in a just, equitable and open scholarly communication landscape.
www.universiteitenvannederland.nl/en/current/events/what-do-we-want-or-not-want-from-publishers
This served as input for the new agreement with publisher Elsevier (2025)
www.universiteitenvannederland.nl/en/current/news/dutch-research-institutions-and-elsevier-announce-new-agreements

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy? (ref.: (ii) 17.j, 23.c)

YES / UNDER DEVELOPMENT / NO *[If yes, please provide details and links as relevant.]*

- One of the tasks assigned to Open Science NL by the Ministry of Education, Culture and Science is to monitor, evaluate and report regularly on the progress of open science in the Netherlands. €400k has been allocated for this purpose in the current Work Programme (2024-2025). A tender has been published in the course of January 2025. Interested parties have been able to express their interest. A selection procedure is currently taking place.
- Regarding open access to scientific publications, since 2016 Universities NL, upon request of the Ministry of Education, Culture and Science, is responsible for the national monitoring framework for open access to scientific publications. UNL has since reported annually on the progress on open access. These percentages have been published in the annual (budget) reports of the Ministry to Parliament.

¹³ www.nwo.nl/en/implementation-guidelines-plan-s

Year	Gold DOAJ OA	Hybrid and non-DOAJ OA	Only green OA	Total OA
2016	13%	20%	9%	42%
2017	17%	23%	10%	50%
2018	19%	24%	11%	54%
2019	21%	29%	12%	62%
2020	25%	35%	13%	73%
2021	28%	39%	15%	82%
2022	30%	39%	20%	89%
2023	30%	44%	18%	92%

Sources: www.universiteitenvannederland.nl/percentages-open-access-publicaties-2016-2023
www.openaccess.nl/en/about-open-access/open-access-monitor

- National funding organisations NWO and ZonMW also developed an open access monitor and report annually on the openness of publications from research funded by both councils. For more than three years over 90% of the articles published with funding are open access. [<https://zenodo.org/records/12685800>]
- NWO and ZonMW have also investigated the availability of research data from research they funded: www.nwo.nl/en/news/sharing-research-data-and-software-how-are-nwo-and-zonmw-funded-projects-doing

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.a](#))

2,23% of the GDP in 2023, according to the Central Bureau of Statistics.

Sources:

- Central Bureau of Statistics: www.cbs.nl/nl-nl/cijfers/detail/84644ENG
- Rathenau Institute: www.rathenau.nl/en/science-figures/investments/how-much-does-netherlands-spend-rd/rd-expenditure-netherlands-sector

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.b](#))

In 2024 98,5% of the Netherlands' population had internet access at home, according to the Central Bureau of Statistics. www.cbs.nl/nl-nl/cijfers/detail/84888NED

3.3 Are there national research and education networks (NRENs) active in the country? (ref.: [\(iii\) 18.c](#))

YES / NO [If yes, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.]

SURF is the National Research and Education Network (NREN) in the Netherlands.

- About SURF: www.surf.nl/en/about
- Members of SURF: In the SURF cooperative, the Dutch universities, universities of applied sciences, VET (MBO) institutions, university medical centres and research institutes work together on groundbreaking IT innovations. Together, they make better and more flexible education and research possible. Most of the education and research institutions in the Netherlands (over 100) are affiliated to the SURF cooperative. www.surf.nl/en/about/members-of-surf

3.4 Are there national or institutional open science infrastructures in your country? (ref.: [\(iii\) 18.d](#), [18.e](#), [18.k](#), [18.l](#), [18.m](#))

YES / UNDER DEVELOPMENT / NO [If yes or under development, please indicate which type(s) of infrastructure.]

- **federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.**
 - **SURF** has a High Performance Computer (HPC), a cloud and data storage. www.surf.nl/en/ / www.surf.nl/en/themes/open-science / www.surf.nl/en/themes/research-infrastructure / www.surf.nl/en/services/hpc-cloud
 - **DANS** is the Dutch national centre of expertise and repository for research data. <https://dans.knaw.nl/en/>

- **4TU.ResearchData** is a data and software repository for science, engineering and design.
<https://data.4tu.nl/>

[We are not sure how to exactly understand the following questions of which many things could be part. Some possible examples are provided.]

- **community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society**
 - the work of the Royal Library (KB) on open science, incl. citizen science and engagement with society: www.kb.nl/en/over-ons/expertises/open-science
 - the work of the University Libraries (UKB) and National Library, e.g. the current research into the establishment of a public platform for presenting scientific works. <https://ukb.nl/en/open-scholarly-communications/>
 - Citizen Science Lab of the University of Leiden www.universiteitleiden.nl/en/citizensciencelab
- **platforms for exchanges and co-creation of knowledge between scientists and society**
 - Citizen Science Nederland network as a national community of practice for CS and participatory research, and the CS-NL Knowledge Platform on <https://citizenscience.nl/> as facilitating knowledge exchange.
 - Citizen Science Lab of the University of Leiden www.universiteitleiden.nl/en/citizensciencelab
 - Platform for open educational resources in higher education <https://edusources.nl/>
- **community-based monitoring and information systems to complement national, regional and global data and information systems.**
 - The CRIS-system : Since the early 1990s, the Dutch research information landscape has shaped by the METIS Current Research Information System (CRIS). In such a system information on research projects, associated personnel, organisations, funding and output is administered which are then used in reports for e.g. management information and research evaluation.
<https://wiki.surfnet.nl/spaces/nlcerif/pages/10862465/About>
 - CWTS-Leiden activities in this field and involvement in the Barcelona Declaration on Open Research Information. www.cwts.nl/news?article=n-t2q294&title=barcelona-declaration-promotes-openness-of-research-information

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

YES / NO

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

If no, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country? YES / NO

[We are not sure how to exactly understand question 3.5 and what to answer here.]

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

YES / NO [*If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science***Fout! Bladwijzer niet gedefinieerd.***, stage of research, accessibility to all the relevant users in the region or internationally).*]

- The Netherlands has recent bilateral national MoU's with e.g. Brazil and South Africa that contains opens science as a topic for (further) cooperation.
- The Netherlands is member, together with fellow-members from the Global South, of various multilateral platforms and organisations that work on that, e.g.
 - GO FAIR VODAN
www.go-fair.org/ , www.go-fair.org/implementation-networks/overview/vodan/
 - EU-Africa cooperation
https://research-and-innovation.ec.europa.eu/system/files/2023-07/ec_rtd_eu-africa-cooperation-in-ri.pdf,
 - Large research infrastructures, such as SKAO
www.skao.int/en/explore/sustainability-skao/513/sustainability-science

- UNESCO Chair at CWTS on Diversity and Inclusion on Global Science
www.cwts.nl/research/unesco-chair-on-diversity-and-inclusion-in-global-science

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: [\(iv\) 19.a, 19.c](#))

YES / NO [If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.]

Open Science NL supports capacity building in higher education and research performing institutions. In the Open Science NL Work programme 2024-2025 in Chapter 1, which is dedicated to Capacity Building, it is stated:

“Investing in capacity building for open science is one of the essential requirements for putting open science into practice. In this document, capacity building refers to the process of enhancing the organisational or community-based knowledge, skills and abilities needed to adopt open science practices effectively. In other words, capacity building refers to investment in people and their skills. Capacity building initiatives provide much needed training and support to navigate the technical, legal, social, cultural and ethical aspects of open science, thereby enabling communities to overcome barriers and embrace new approaches. Ultimately, capacity building strengthens the foundations of sustainable open science ecosystems, which encompass not only specific research outputs (publications, data software and non-traditional research output, such as creative works, policy reports, etc.), but also research processes, infrastructures and research cultures. Over the past few years NWO has already been investing in capacity building at the institutional level, mainly through calls to support digital competence centres: Local Digital Competence Centres (LDCCs) and Thematic Digital Competence Centres (TDCCs). The four funding programmes proposed here will build on these earlier initiatives to further stimulate the capacity building for open science:

Over the past few years NWO has already been investing in capacity building at the institutional level, mainly through calls to support digital competence centres: Local Digital Competence Centres (LDCCs)⁸ and Thematic Digital Competence Centres (TDCCs). The four funding programmes proposed here will build on these earlier initiatives to further stimulate the capacity building for open science:

1.1. National Training Platform for Research Data Professionals – providing national-level training and community support for aspiring data professionals.

1.2. Strengthening local and thematic Digital Competence Centres – increasing software training capacity and investing in interoperability of research data.

1.3. Citizen Science Hubs – fostering the creation of regional hubs with citizen science and societal engagement expertise. –

1.4. Enabling and strengthening institutional open publishing – supporting networks and building expertise in non-profit, university-based publishing venues.

Overall budget in 2024-2025: €22.4M ”

Source: www.openscience.nl/en/news/open-science-nl-presents-work-programme-for-2024-and-2025 pages 7-9

- See on the website of the various higher education and research performing institutions mentioned under question 2.5. for specific examples at institutional level.

4.2 Have capacity building programmes for policy makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

YES / NO [If yes, please provide more information and links as relevant. If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.]

Not so much capacity building programmes dedicated specifically to policy makers, however a huge effort is being made by institutions to build capacity (and knowledge) through networks and communities (e.g., library working groups, RDM networks and Open Science Communities and Citizen Science-NL).

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

YES / **PARTLY** / NO

Such a framework could possibly be something for UNESCO to (help) develop.

In the Netherlands, at institutional level most universities offer courses, primarily at PhD- and research MA-level. Some examples:

- Erasmus University: Course on Open Science and Research Transparency
www.eur.nl/en/egsh/course/open-science-and-research-transparency
- Utrecht University- www.uu.nl/en/education/graduate-school-of-life-sciences/phd/starting-your-phd/responsibilities-of-phd-candidates/open-science
- TU Delft:
<https://online-learning.tudelft.nl/courses/open-science-sharing-your-research-with-the-world/>
- Leiden University - Open Science, A practical guide for early career researchers (2023)
www.library.universiteitleiden.nl/news/2023/06/now-available-open-science-a-practical-guide-for-early-career-researchers

Open Science NL provides practical information and inspiration to knowledge institutions and researchers.
www.openscience.nl/praktische-informatie-inspiratie

At national level such a framework does not exist, as the Netherlands does not know a national curriculum.

4.4 Are open educational resources as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

YES/ PARTLY/ NO

Yes (e.g. / via / by):

- eduSources, the place for open educational resources in higher education
<https://edusources.nl/>
- NPULS
<https://npuls.nl/en/digital-educational-resources/>
- Acceleration Plan Educational Innovation with ICT
www.versnellingsplan.nl/en/Kennisbank/national-approach-digital-open-educational-resources/

Some examples at institutional level:

- University of Utrecht, open science and open education
www.uu.nl/en/research/open-science/knowledge-center/good-practices-faq/open-education
- Wageningen University & Research, open science and open education
www.wur.nl/en/about-wur/our-values/finding-answers-together/open-science-education.htm
- Erasmus University Rotterdam
www.eur.nl/en/library/education-support/lecturer/open-educational-resources/events

4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

YES / NO [If yes, please provide more information and links as relevant.]

This is being done e.g. by the National Expert Centre for Science and Society (NEWS) that was established in 2024. <https://wetenschapensamenleving.nl/index.php/over-ons-2/>

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

YES / NO [If yes, please provide more information and links as relevant.]

Open Science NL has in December 2024 awarded €1.14 million to 24 research institutions in order to embed open science in recruitment and promotion criteria. A further €150k was awarded to support national coordination:

www.openscience.nl/en/news/knowledge-institutions-take-action-on-open-science-in-recruitment-and-promotion-policies

Dutch universities, university medical centers, research institutes and research funders have collectively launched the national Recognition & Rewards Programme in 2019. The programme aims to achieve a new balance in the way academics are recognized and rewarded. One of the five main ambitions is to stimulate all aspects of open science in the assessment and evaluation of academics. In 2023, the public research institutions set the goal to (further) clarify how activities relating to open science and open education will be considered and/or prioritised as a topic of discussion in the development, assessment, appointment, and promotion of staff. Within the institutions, teams are actively working on and with the principles of open science and FAIRification. The institutions aim to make sure employees' efforts in open science will be part of the evaluation and annual appraisal interview. The Recognition & Rewards Programme is co-funded by the Dutch Ministry of Education, Culture and Science (2022-2026).

See: <https://recognitionrewards.nl/about/about-the-programme/>
<https://recognitionrewards.nl/about/roadmap/>

5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

YES / NO [If yes, please provide details and links as relevant.]

All aspects of open science are promoted within the national Recognition and Rewards programme. (See the answer to question 5.2). Some institutions have formally adopted the principles of open science into evaluation procedures. For example:

- At Maastricht University, impact & open science is a mandatory part of assessment of all senior academics (R3-R4).
www.maastrichtuniversity.nl/file/assistantassociateandfullprofessorprofileinclengpdf
- At VU Amsterdam, open science, like academic integrity, is conditional to performance assessments and promotion.
https://assets-us-01.kc-usercontent.com/d8b6f1f5-816c-005b-1dc1-e363dd7ce9a5/de3480c3-3962-4a3e-a934-b61fe6deb90e/Academic%20Careerpath%202023%20VU_EN.pdf
- At Utrecht University, the promotion profile for Professor 1 (R4) requires the Professor 1 to “use their own position to actively promote Open Science”
www.uu.nl/sites/default/files/Promotion-Profile-and-Procedure-Prof-1-EN.pdf

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20.h)

YES / NO [If yes, for each initiative, please provide more information, including on the stakeholdersⁱ that are involved or lead the initiative, and links as relevant.]

- All aspects of open science are promoted within the national Recognition and Rewards programme. See the answer to questions 5.2 and 5.3.
- Various Open Science Awards, e.g.:
<https://opensciencefestival.nl/en/news/who-will-be-the-winner-of-the-first-leo-waaijers-open-science-award>
<https://vu.nl/nl/nieuws/2023/vu-open-science-challenge-award-2023-voor-8-projecten>
www.hva.nl/subsites/nl/hva-bibliotheek/nieuws/2024/04/oproep-voor-open-science-awards-2024.html
www.medewerkers.universiteitleiden.nl/nieuws/2022/11/open-science-prijs-voor-anne-urai-en-collegas
- Research Data Netherlands (RDNL) has been organising every two years the Dutch Data Prize. The prize is awarded to an individual or a team that makes research data FAIR: Findable, Accessible, Interoperable and Reusable. The Dutch Data Prize is awarded in three categories: Social Sciences and Humanities; Natural and Engineering Sciences; Life Sciences and Health.
- CWTS overview article ‘Monitoring Open Science in The Netherlands’ (2025), including references to the UNESCO Open Science Recommendation: www.leidenmadtrics.nl/articles/monitoring-open-science-in-the-netherlands

5.4 Have there been any unintended negative consequences of open science practices reported in your country? (ref.: (v) 20)

YES / NO *[If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.]*

No reported unintended negative consequences. However, the (international) shift from 'pay to read' to 'pay to publish' business models, including risen Article Processing Costs may be an example here. For the Global South this shift is even harder to bear. The UNESCO Chair at CWTS on Diversity and Inclusion in Global Science is -among other things- looking into this.

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

YES, at both *national and institutional levels*, e.g.:

- **Dutch Research Agenda:** Citizens are at the heart of creating and renewing the Dutch Research Agenda. The link between society and science is a key element of the NWA programme. Governments, researchers, civil society organisations and citizens work together to develop and use knowledge to achieve scientific and societal impact. And through science communication, knowledge is shared in society to increase engagement and trust in science.
www.nwo.nl/en/researchprogrammes/dutch-research-agenda-nwa
<https://wetenschapsagenda.nl/>
- **Open Science NL:** several of its funding instruments are aiming at this. www.openscience.nl/ OS NL works closely together with and receives input from the institutional level.
- **Citizen Science NL** as the national network (<https://citizenscience.nl/>) and **Citizen Science Hubs** at institutional level aiming at participatory activities. (www.openscience.nl/en/researchprogrammes/citizen-science-hubs)

If yes, please indicate which of the recommended actions they address and provide more information and links as relevant:

- ***promotion of pre-prints and exploration of open peer-review practices:***
 - this is being addressed ad hoc at different institutional levels, see for example: www.universiteitleiden.nl/en/news/2023/10/posting-preprints-there-is-no-reason-not-to
www.staff.universiteitleiden.nl/news/2024/02/open-and-free-peer-review-heres-how
 - or through guidance, see: <https://zenodo.org/records/5600535>
 - quite a few institutions are actively supporting infrastructure for preprints and open peer reviews. E.g., all universities and NWO are supporting ArXiv.
 - there is currently no policy (or mandate) for preprints in the Netherlands, but this is being discussed.
- ***valuing and sharing of negative scientific results and associated data:***
 - this is being addressed in different communities. There is even a journal originated from the Netherlands, addressing this topic specifically: <https://journal.trialanderror.org/>
- ***participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific project:***
 - **Dutch Research Agenda:** it is a participatory method in itself.
 - **Citizen Science NL** (citizenscience.nl) and its knowledge platform
- ***development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas.***
This is incorporated in the Dutch Research Agenda.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

NWO has been providing financial support for the key international infrastructures for open science:
www.nwo.nl/en/open-science-infrastructure-support

In recent years, several university libraries have set up dedicated funds for supporting (inter)national infrastructures to support open science (e.g., www.uu.nl/en/university-library/advice-support-to/researchers/publishing-support/open-access/open-access-in-practice/diamond-open-access). Strengthening the coordination and developing collective funding mechanisms is currently underway.

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

YES / NO *If yes, please provide links as relevant and more information.*

In several of the letters to Parliament mentioned under question 2.2. it is stated that the Netherlands will continue to play an active role in cross-border (multi-stakeholder) collaboration on open science. The Netherlands does so e.g.:

- via its membership of, or affiliation with, various multilateral organisations, such as the European Union, the OECD, UNESCO (e.g. the Netherlands in 2024 donated € 100.000,- to UNESCO for the implementation of the Recommendation on Open Science) and the G20.
- via bilateral (high-level) contacts, including in the follow-up to Memoranda of Understanding in the field of science and research that were signed in recent years with several countries, in the Global North as well as the Global South, in which open science is mentioned as a topic for (further) cooperation.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science?

(ref.: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/page=21>)

YES / NO [*If yes, please provide links as relevant and more information.*]

NWO is financially supporting the Global Research Initiative on Open Science (GRIOS - <https://grios.org/>), which aims to review the existing evidence and ongoing research on open science.

8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES / NO (*If yes, please provide more information.*)

It is the ambition of the Dutch government that open science will be the norm in scientific research in the Netherlands by the year 2032. Although, due to upcoming general budget cuts, there will be less budget available for the transition to open science than initially anticipated, the ambition remains unchanged.
